Student Dropout Prediction: A Machine Learning Analysis

DSAA2011 (L01) Course Project — HKUST(GZ) 2026 Spring

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Abstract

We analyze the UCI Student Dropout and Academic Success dataset using a complete machine learning pipeline covering preprocessing, t-SNE visualization, clustering, supervised classification, model evaluation, and open-ended model exploration. The final engineered top-40 feature space contains strong predictive signal, especially from academic performance features such as semester approval rates and grades. Unsupervised analysis shows that the three outcomes are not naturally separable, with the Enrolled class lying between Dropout and Graduate. Among mandatory baseline classifiers, Logistic Regression is the strongest and most stable baseline. In open-ended exploration, a tuned Random Forest achieves the best held-out test macro F1 of 0.7096, only slightly above Logistic Regression at 0.7079, so the final result should be interpreted as a marginal predictive improvement rather than a decisive model gap.

1 Introduction

The Student Dropout and Academic Success dataset from the UCI Machine Learning Repository [1] contains 4,424 instances of students enrolled in various undergraduate programs at a Portuguese higher education institution. Each instance is described by 36 features covering demographics, socioeconomic background, and academic performance across the first and second semesters.

Our objectives are: (1) explore data structure through visualization and clustering; (2) train and evaluate supervised classifiers; (3) improve models via validation-driven tuning; (4) investigate feature importance and ensemble methods.

The report follows an evidence-first structure. We first construct an engineered top-40 feature space, then use t-SNE and clustering to examine whether student outcomes form natural groups. Because Enrolled students remain structurally intermediate between Dropout and Graduate, the supervised sections evaluate whether label-informed decision boundaries can better separate the outcomes. The strongest mandatory baseline is Logistic Regression, while the open-ended model comparison selects tuned Random Forest by held-out test macro F1.

2 Mandatory Tasks

2.1 Data Preprocessing

The dataset contains 36 raw features that are categorized into four types: *nominal* (8 features), *binary* (8 features), *ordinal* (13 features), and *continuous* (7 features). It also includes a three-class target variable: `Dropout` (32.1%), `Enrolled` (18.0%), and `Graduate` (49.9%). We performed a comprehensive data preprocessing pipeline as outlined below.

Missing values and outliers. As stated in the dataset description, the dataset is clean with no missing values, as confirmed by a complete null value check across all features.

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EDA for nominal features. We inspected the distribution and top categories of nominal features. Except for feature `Course`, most nominal features are heavily skewed and the top four categories can cover over 80% of the data. This informed our decision to apply out-of-fold (OOF) target encoding for these features, as one-hot encoding would lead to high dimensionality and sparsity.

EDA for numerical correlations. We first sorted the 38 numerical features by their absolute Pearson correlation with the target variable. The top 15 features with $|r| \geq 0.09$ along with the target variable were visualized in a correlation heatmap (Figure 1).

1. Curricular units between 1st and 2nd, or between grade and admission grade, show strong correlations with each other and with the target variable. These features are likely important predictors of dropout risk. 2. Economic features (e.g., 'tuition fees', 'debtor', 'displaced') show moderate correlations with each other and with the target variable, suggesting that financial factors may also play a role in dropout risk. 3. Other factors such as 'age', 'gender', and 'scholarship holder' show weaker but notable correlations with the target variable, indicating that demographic factors may also contribute to dropout risk.

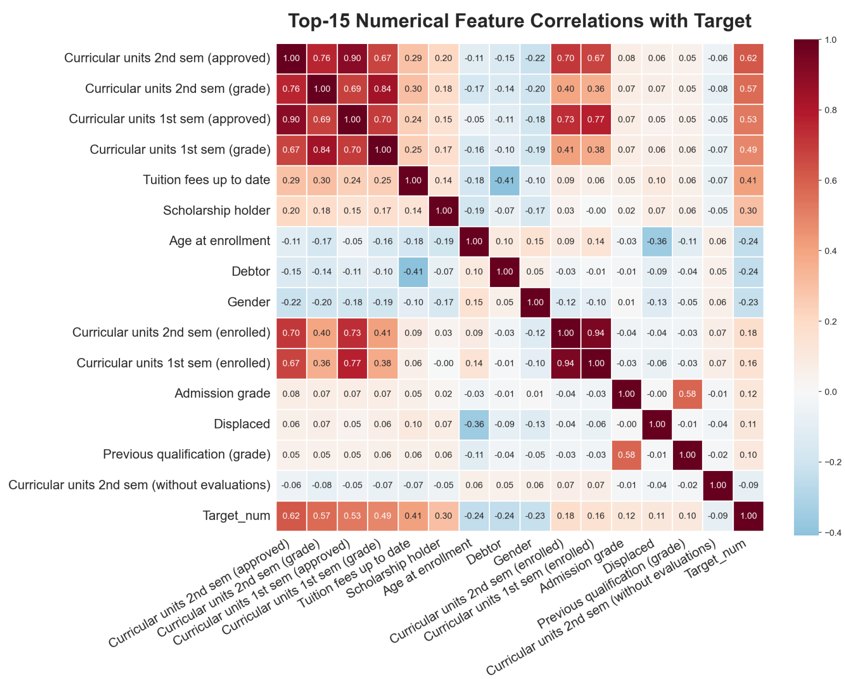

Figure 1: Top-15 feature correlations with the target variable.

EDA for visualizing important features. To gain a better understanding of the curricular units that show strong correlations with the target variable, we visualized the distributions of the four features (1st/2nd semester, grade/approved units) as histograms with trendlines (Figure 2). These visualizations confirm that higher values in these features are associated with the Graduate class, while lower values are more common among Dropout students. Enrolled students show intermediate distributions, consistent with their transitional academic status.

Feature engineering and selection. We perform feature engineering to create 25 new domain-specific features based on the original 36 features, resulting in a total of 61 candidate features.

Data splitting and target encoding. The dataset was split into training (70%), validation (15%), and test (15%) sets using stratified sampling to maintain class proportions. Then OOF (Out-of-Fold) target encoding was applied to the nominal features using the training set to compute the mean target value for each category, which was then applied to the validation and test sets. This approach prevents data leakage while allowing us to capture the predictive signal from categorical features.

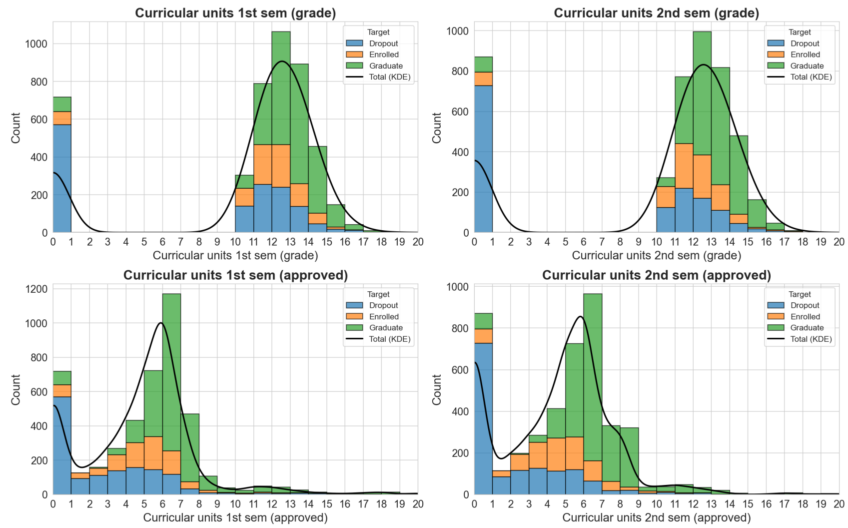

Figure 2: Distributions of top curricular unit features across target classes.

Feature standardization, selection, and output. Finally, we standardized the numerical features to have zero mean and unit variance. We then applied a feature selection method by training a Random Forest classifier on the training set and using its feature importance scores to select the top 40 features for modeling. The final preprocessed datasets were saved for subsequent analysis.

2.2 Data Visualization (t-SNE)

We applied t-SNE to the engineered feature dataset to project the high-dimensional feature space into lower dimensions for visualization.

A systematic pattern analysis across four perplexity values (Figure 3) highlights the algorithm's sensitivity to local versus global structure. At a low perplexity ($p = 5$), the projection is severely fragmented, which obscures meaningful global patterns. Conversely, a high perplexity ($p = 50, 100$) over-smooths the global structure, causing a loss of local detail while significantly increasing computational expenditure.

Perplexity=30 achieves the optimal balance. The 2D projection (Figure 3) reveals that Graduate students form a relatively compact region, while Dropout students cluster separately. Enrolled students are scattered between both groups, which is consistent with their transitional academic status. Furthermore, extending the projection from 2D to 3D (Figure 4) provides additional spatial volume, further reducing the overlap between the distinct Graduate and Dropout clusters.

2.3 Clustering Analysis

Three algorithms were compared: K-Means, Agglomerative Hierarchical Clustering (ward, complete, average, single linkage), and Gaussian Mixture Models (GMM).

Exploratory Experiments: Before settling on the final algorithm set, we conducted several exploratory experiments. DBSCAN was initially considered for its density-based approach and ability to detect arbitrary-shaped clusters without pre-specifying K. However, on the engineered feature space, DBSCAN didn't find valid clusters: the grid search over $\epsilon \in \{0.1, 0.5, 1.0, 2.0, 5.0\}$ and $\min\_samples \in \{3, 5, 10\}$ classified 94–100% of points as noise. t-SNE was also explored for dimension reduction prior to clustering. However, for clustering visualization, PCA was preferred because it provides a stable linear projection more suitable for comparing clustering results, whereas t-SNE is stochastic and primarily intended for nonlinear visualization rather than distance-preserving projection.

K-Means Clustering: We applied K-Means with K=3 based on the known 3-class structure of the target variable.

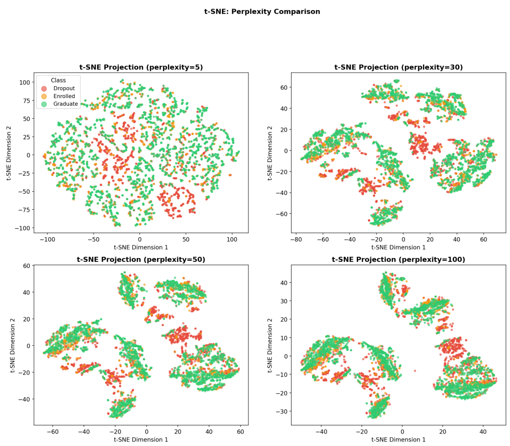

Figure 3: t-SNE 2D projections at perplexity = 5, 30, 50, 100.

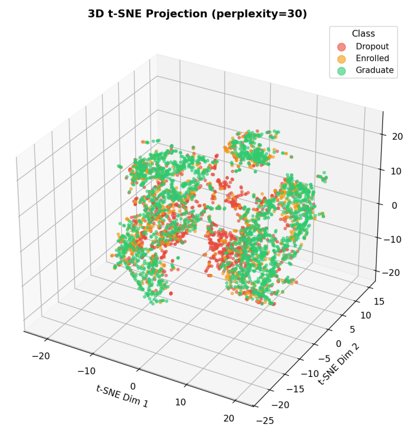

Figure 4: t-SNE 3D projection at perplexity=30.

K-Means produced reasonably balanced cluster sizes, but the alignment with true class labels was weak (ARI=0.003) (Table 2). All three clusters are dominated by Graduate students (47–52% each) (Table 3), indicating severe overlap between the three student categories in the engineered feature space.

Hierarchical Clustering: The dendrogram for Ward linkage (Figure 5) shows a relatively flat hierarchical structure, which also supports K-Means' spherical cluster assumption. Single and average linkages produced degenerate partitions with near-zero ARIs, while Ward and complete linkages generated more balanced structures (Table 1).

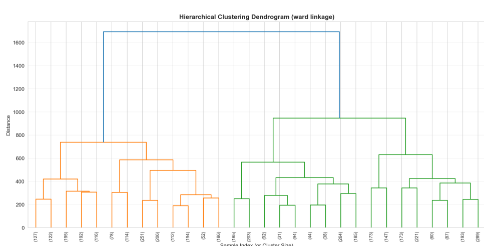

Figure 5: Hierarchical clustering dendrogram (Ward linkage).

Table 1: Linkage methods comparison (K=3).

Method	Silhouette	CH Index	DB Index	ARI
Ward	0.2314	1226.76	1.7040	0.0023
Complete	0.1291	420.47	2.1586	0.0086
Average	0.3166	37.54	1.1090	−0.0006
Single	0.3973	5.08	0.4261	−0.0003

GMM Analysis: GMM models data as a mixture of Gaussians, providing soft probabilistic clustering rather than hard boundaries. We set K=3 to match the true class count. Despite a lower Silhouette score (0.033), GMM achieved substantially higher ARI (0.174) than both K-Means and Hierarchical clustering (Table 2), indicating its soft-assignment mechanism captures genuine structural patterns missed by centroid-based methods.

Overall Algorithm Comparison: Table 2 summarizes the performance of all three clustering algorithms across six evaluation metrics. GMM achieves the highest ARI (0.174) and NMI (0.118), outperforming K-Means and Hierarchical clustering on alignment with true labels. But GMM has a lower Silhouette score and higher DB Index, reflecting the overlapping nature of probabilistic cluster boundaries. Table 3 shows the dominant true class per cluster for each algorithm, which reflects that only GMM produces dropout-dominant clusters (Clusters 1 and 2), while K-Means and Hierarchical clustering are dominated by Graduate across all clusters. Figure 9 visualizes the clustering results of all three algorithms on PCA 2D projection, where PCA reduces the 40-dimensional engineered feature space to two dimensions for visualization purposes.

Table 2: Clustering algorithm performance comparison.

Algorithm	K	Silhouette	CH Index	DB Index	ARI	NMI
K-Means	3	0.2421	1304.47	1.5809	0.0033	0.0054
Hierarchical (Ward)	3	0.2314	1226.76	1.7040	0.0023	0.0043
GMM	3	0.0328	217.01	4.7831	0.1742	0.1176

Table 3: Dominant true class per cluster for each algorithm.

Algorithm	Cluster 0	Cluster 1	Cluster 2
K-Means	Graduate (50.5%)	Graduate (47.8%)	Graduate (51.5%)
Hierarchical (Ward)	Graduate (50.4%)	Graduate (48.0%)	Graduate (51.4%)
GMM	Graduate (67.8%)	Dropout (59.6%)	Dropout (52.5%)

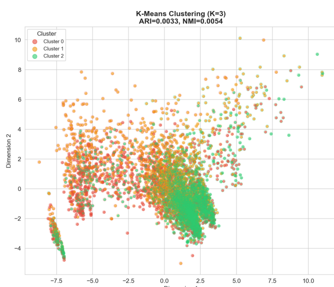

Figure 6: K-Means (K=3)

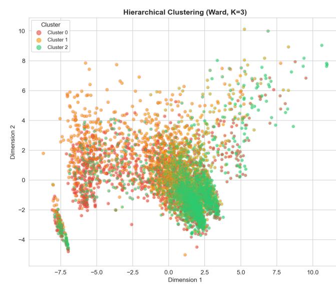

Figure 7: Hierarchical (Ward, K=3)

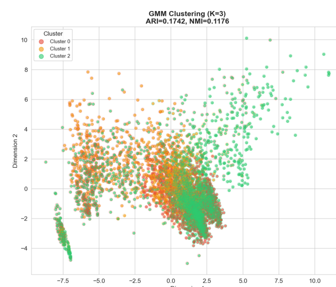

Figure 8: GMM (K=3)

Figure 9: Cluster visualization on PCA 2D projection. red - Dropout, orange - Enrolled, green - Graduate.

Best clustering: GMM. Based on analysis above, GMM is selected as the best clustering algorithm, indicating that the probabilistic soft-assignment mechanism captures more structural patterns that centroid-based methods miss.

2.4 Prediction: Training and Testing

The clustering results suggest that the three student outcomes are not naturally separated in the engineered feature space. We therefore train supervised classifiers on the top-40 engineered features to test whether label-informed decision boundaries can better distinguish Dropout, Enrolled, and Graduate students.

Data preparation. We used the engineered feature dataset (40 features after OOF target encoding and top-40 selection), split into train (3,096), validation (664), and test (664) sets. The class distribution in the training set is: Dropout 32.1%, Enrolled 17.9%, Graduate 49.9%.

Decision Tree (DT), Logistic Regression (LR), and SVM with an RBF kernel were selected as mandatory baseline classifiers because they represent complementary assumptions. DT captures non-linear splits but can overfit, LR provides a stable and interpretable linear baseline, and RBF-SVM can model non-linear boundaries but is sensitive to hyperparameters and class overlap.

We trained each classifier using default configurations from `prediction.py`: Decision Tree (`max_depth=10, class_weight='balanced'`), Logistic Regression (`solver='saga', class_weight='balanced'`), and SVM with RBF kernel (`class_weight='balanced'`). All models were evaluated on training, validation, test, and entire sets.

Multi-set evaluation. Table 4 reports accuracy and macro F1 across all evaluation sets. DT has the largest train-test gap, indicating overfitting. LR has the strongest held-out test macro F1 among the mandatory baselines and remains stable across train, validation, and test splits. Default RBF-SVM performs weakest by macro F1, suggesting that the default margin and kernel settings are not enough for the overlapping class structure.

Table 4: Baseline model performance (macro F1 across splits and test accuracy).

Model	Train F1	Val F1	Test F1	Test Acc.
Decision Tree	0.8876	0.6693	0.6769	0.7139
Logistic Reg.	0.7156	0.7070	0.7079	0.7410
SVM (RBF)	0.6394	0.6166	0.6449	0.6867

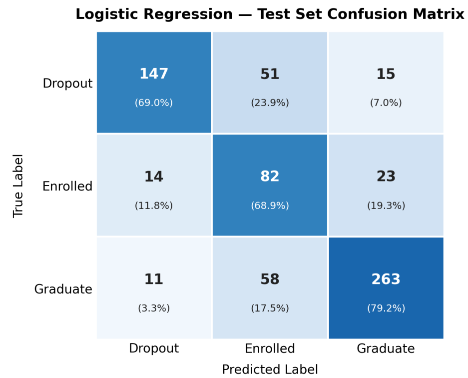

Figure 10: Logistic Regression confusion matrix on the test set.

The per-class analysis shows that all models achieve the highest recall for Graduate and the lowest for Enrolled. This is consistent with the class imbalance and the transitional nature of enrolled students, who share feature characteristics with both Dropout and Graduate.

Error analysis. The confusion matrix (Figure 10) reveals the dominant error pattern: Graduate students are correctly identified with the highest recall (79.2%), while Enrolled students show the lowest (68.9%), with 11.8% misclassified as Dropout and 19.3% as Graduate. This confirms that Enrolled does not occupy a distinct decision region. The pattern aligns with the t-SNE visualization (Section 2.2) and clustering results (Section 2.3), where Enrolled points lie in the overlapping zone between the two more separable outcomes. The structural overlap between classes, beyond class imbalance alone, is a major source of classification difficulty.

2.5 Evaluation and Model Choice

We evaluate the baseline models from three complementary perspectives: (1) held-out test performance via precision, recall, and F1; (2) ranking ability through One-vs-Rest ROC-AUC; and (3) robustness under cross-validation and hyperparameter variation. This section compares and tunes the three mandatory baseline models from Section 2.4. The overall final model choice will be made in the open-ended exploration section, where ensemble methods are evaluated.

We retrained the baseline models with explicit configurations and computed detailed precision, recall, and F1 metrics. We then applied One-vs-Rest ROC curves, cross-validation, validation curves, learning curves, and GridSearchCV hyperparameter tuning.

Detailed metrics. Table 5 reports the full set of evaluation metrics on the test set. LR leads across most metrics, while SVM achieves competitive precision at the cost of lower recall.

Table 5: Detailed per-model metrics on the test set.

Model	Acc.	Prec. (macro)	Recall (macro)	F1 (macro)
Decision Tree	0.7139	0.6871	0.6924	0.6769
Logistic Reg.	0.7410	0.7192	0.7238	0.7079
SVM (RBF)	0.6867	0.6886	0.6622	0.6449

Table 6: Per-class and Macro AUC.

Model	Dropout	Enrolled	Graduate	Macro
DT	0.800	0.704	0.849	0.784
LR	**0.913**	**0.825**	**0.931**	**0.890**
SVM	0.895	0.796	0.907	0.866

ROC curves and AUC. Figure 11 compares macro-average ROC curves for all three baselines. LR achieves the highest macro AUC (0.890), followed by SVM (0.866) and DT (0.784).

Cross-validation. 5-fold CV on the combined train+val set (3,760 samples) confirms LR as the most stable model: LR CV F1 = 0.7028 ± 0.0073, DT = 0.6533 ± 0.0117, SVM = 0.6342 ± 0.0153.

Validation curve analysis reveals model-specific sensitivity to hyperparameters. Figure 12 shows the two most informative curves: DT max_depth exhibits classic overfitting (training score rises while validation drops), while SVM C shows that increasing regularization flexibility steadily improves both training and validation performance.

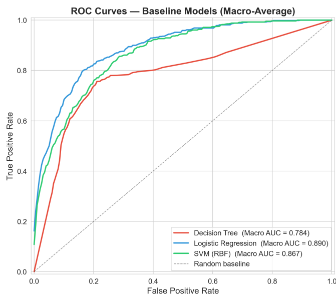

Figure 11: Macro-average ROC curves for all baseline models.

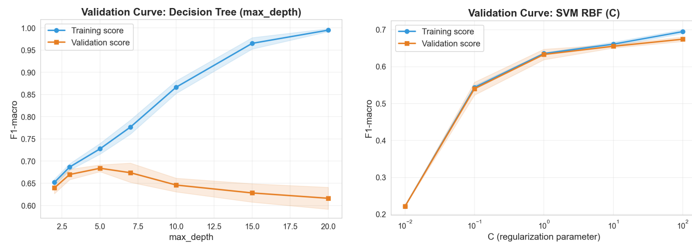

Figure 12: Validation curves: DT max_depth (left) shows overfitting; SVM C (right) shows improvement with larger C.

LR is relatively insensitive to its regularization parameter C, suggesting the linear assumption is the primary bottleneck. SVM gamma (not shown) causes severe overfitting at high values; the optimal C and gamma must be tuned jointly.

Learning curves confirm the validation curve observations: DT exhibits persistent overfitting regardless of training set size, LR has nearly converged with the current data, and SVM's validation score continues to rise, suggesting it could benefit from additional training samples.

Hyperparameter tuning via GridSearchCV produced the best configurations:

DT:	`max_depth=5, min_samples_split=2, min_samples_leaf=1`	(CV F1=0.6837)
LR:	`C=1, penalty='l1', solver='saga'`	(CV F1=0.7075)
SVM:	`C=100, gamma='scale'`	(CV F1=0.6782)

Table 7: Baseline vs. tuned (F1 macro).

Model	Baseline	Tuned	Δ
DT	0.6769	0.6731	-0.0038
LR	0.7079	0.7037	-0.0041
SVM	0.6449	0.6808	$+0.0359$

SVM benefits most from hyperparameter tuning, improving held-out macro F1 from 0.6449 to 0.6808 after increasing model flexibility with a larger C. DT and LR do not improve on the held-out test set after tuning, even though cross-validation helps identify more regularized configurations. This suggests that LR's main limitation is not regularization strength but the linear decision boundary, while DT remains vulnerable to overfitting. The tuned confusion matrices confirm that the Enrolled class remains the most difficult to predict across all configurations.

Baseline summary. Within the mandatory baseline models, LR is the strongest and most stable choice, with the highest test macro F1, macro AUC, and cross-validation stability. DT is interpretable but overfits, and SVM benefits substantially from tuning but remains below LR. Across confusion matrices and ROC curves, Enrolled remains the hardest class because it is both underrepresented and structurally intermediate between Dropout and Graduate. These baseline findings motivate the open-ended exploration of ensemble models rather than ending the analysis with a baseline deployment recommendation.

3 Open-ended Exploration

3.1 Ensemble Methods

The baseline results motivate ensemble models because the data are neither cleanly linearly separable nor well captured by a single decision tree. Random Forest (RF) and Gradient Boosting (GB) can model non-linear feature interactions while reducing the instability of individual trees. Table 8 compares default and tuned ensemble models on the held-out test set.

Table 8: Ensemble model performance on the test set.

Model	Acc.	Prec. (macro)	Recall (macro)	F1 (macro)	F1 (wtd)
RF (default)	0.7696	0.7146	0.6805	0.6881	0.7552
GradientBoosting (default)	0.7575	0.6946	0.6883	0.6910	0.7559
RF (tuned)	0.7515	0.7151	0.7163	0.7096	0.7597
GradientBoosting (tuned)	0.7726	0.7131	0.6965	0.7027	0.7657

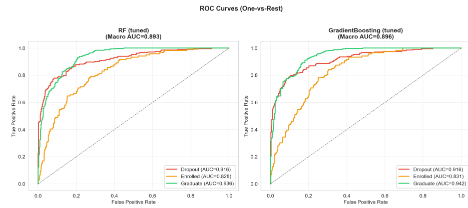

Figure 13: ROC curves for tuned RF and GB (One-vs-Rest).

Tuned RF achieves the highest macro F1 among ensemble models, while tuned GB achieves the highest accuracy. Because the dataset is imbalanced and Enrolled is the hardest class, macro F1 is more appropriate than accuracy for final model selection. Under this criterion, tuned RF is preferred, but the margin over LR is small. The ROC analysis (Figure 13) shows that tuned RF achieves a macro AUC of 0.893, compared with LR's macro AUC of 0.890, confirming that both ensemble models maintain strong ranking ability across all three classes.

3.2 Class Imbalance Handling

Table 9 compares per-class Enrolled F1 with and without balanced class weighting across the three mandatory baseline models.

Table 9: Effect of class weighting on Enrolled F1 score.

Model	Enrolled F1 (default)	Enrolled F1 (balanced)	Δ
Decision Tree	0.4034	0.5032	+0.0999
Logistic Reg.	0.4593	0.5290	+0.0697
SVM (RBF)	0.0000	0.4748	+0.4748

The effect is most dramatic for SVM, which completely ignores the Enrolled class without class weighting (F1 = 0.0000). This indicates that accuracy-oriented training can collapse toward the majority classes when the loss function is not adjusted for imbalance. Balanced weighting improves minority-class recognition across all models, but the remaining Enrolled errors show that imbalance is only one part of the problem; structural overlap with Dropout and Graduate is the other.

3.3 Cross-Validation Comparison

A unified 5-fold cross-validation (Figure 14) compared all five model classes on the combined train+val set:

Table 10: 5-fold CV F1 (macro) results.

Model	CV F1 (Mean $\pm$ Std)	Note
RF	0.7108 ± 0.0187	Highest mean
GB	0.7057 ± 0.0096	Strong ensemble baseline
LR	0.7028 ± 0.0073	Most interpretable strong baseline
DT	0.6533 ± 0.0117	Overfitting risk
SVM	0.6342 ± 0.0153	Weak default setting

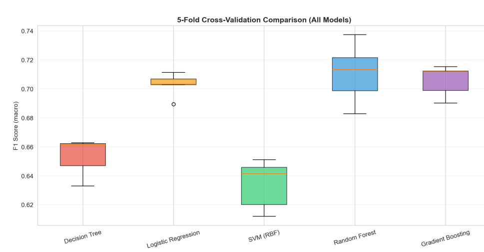

Figure 14: 5-fold CV comparison across all models.

RF achieves the highest mean CV macro F1 (0.7108), while GB provides a strong ensemble baseline with the lowest variance (std = 0.0096). LR remains highly competitive and the most stable individual model (std = 0.0073), reinforcing the conclusion that the ensemble advantage is modest rather than decisive.

3.4 Model Interpretation and Feature Ablation

Feature importance scores from the tuned Random Forest (Figure 15) confirm that academic performance metrics dominate prediction. The top-10 features are led by `overall_approval_rate` and `approval_rate_2nd`, followed by curricular unit metrics and semester grades.

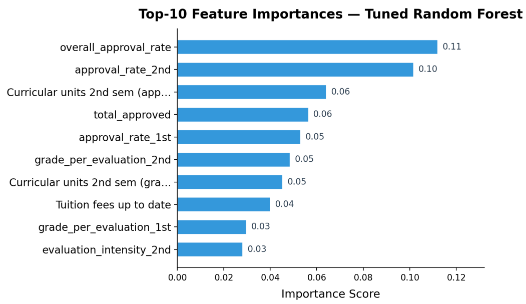

Figure 15: Top-10 feature importances from the tuned Random Forest.

Table 11: Feature group ablation (5-fold CV, tuned RF).

Feature Group	n	CV F1 (Mean $\pm$ Std)
Academic only	20	0.6712 ± 0.0128
Socioeconomic only	20	0.5604 ± 0.0244
All top-40	40	0.7173 ± 0.0094

Feature group ablation. To quantify the relative contribution of different feature types, we partitioned the top-40 features into academic features (20 features: approval rates, grades, curricular unit metrics) and socioeconomic/demographic features (20 features: parental qualifications, economic indicators, application metadata). Table 11 shows the 5-fold CV results using the tuned RF.

Academic features alone account for the majority of predictive signal (F1 = 0.671), while socioeconomic features alone are substantially weaker (F1 = 0.560) with higher variance. Combining both groups yields the best and most stable performance (F1 = 0.717), confirming that secondary features provide complementary discriminative information beyond academic records alone.

Feature count and interaction experiments. We further tested how many features are needed and whether manual nonlinear expansion helps. Feature count experiments (10/20/30/40 selected features, 5-fold CV) show that top-40 achieves the best mean macro F1 (0.699), while top-10 drops to 0.655, confirming that the strongest academic features still need secondary socioeconomic information. We also tested pairwise polynomial interaction features among the top-10 base features (expanding from 40 to 85 dimensions), but interactions did not improve CV F1 ($0.695 \rightarrow 0.686$), suggesting that tree-based ensembles already capture the relevant nonlinear combinations internally. We therefore retain the original top-40 feature space.

Note: The ablation experiments above use controlled settings specific to each comparison (e.g., default vs. tuned RF, different CV folds); absolute F1 values should be interpreted within each experiment rather than compared directly across tables.

3.5 Comprehensive Model Comparison

Table 12 presents the full comparison of all seven model configurations on the test set.

Using held-out test macro F1 as the selection criterion, tuned RF is the final model with macro F1 = 0.7096. This result should be interpreted cautiously because LR is very close at 0.7079 and remains simpler and more interpretable. The practical conclusion is therefore not that RF dominates all baselines, but that ensemble tuning provides a small additional gain while preserving the broader finding that Enrolled is the primary classification challenge.

Per-class analysis of the final model. Table 13 compares per-class F1 for tuned RF and LR, both trained on the combined train+validation set for fair comparison. Tuned RF improves F1 across all three classes, with the largest gain on Graduate (+0.009) and modest gains on Dropout (+0.006) and Enrolled (+0.005). Enrolled remains the weakest class for both models (F1 $\approx$ 0.52), confirming that the primary prediction difficulty is structural overlap rather than model choice.

Statistical significance. A bootstrap analysis (10,000 iterations on the test set) yields a 95% confidence interval for the macro F1 difference (RF − LR) of $[-0.018, +0.032]$ with $p = 0.30$, indicating that the observed advantage of tuned RF is not statistically significant. This reinforces the practical

Table 12: Comprehensive comparison of all models on the test set.

Model	Accuracy	F1 (macro)	F1 (weighted)
Decision Tree	0.7139	0.6769	0.7276
Logistic Regression	0.7410	0.7079	0.7553
SVM (RBF)	0.6867	0.6449	0.7005
RF (default)	0.7696	0.6881	0.7552
GradientBoosting (default)	0.7575	0.6910	0.7559
RF (tuned)	0.7515	**0.7096**	0.7597
GradientBoosting (tuned)	**0.7726**	0.7027	**0.7657**

Table 13: Per-class F1 comparison: tuned RF vs. LR (both trained on train+val).

Class	RF F1	LR F1	Δ
Dropout	0.767	0.761	$+0.006$
Enrolled	0.521	0.516	$+0.005$
Graduate	0.840	0.831	$+0.009$

conclusion: the two models are effectively equivalent in predictive power, and the choice between them involves a trade-off between marginal F1 gain and model interpretability.

4 Conclusion

Key findings:

- The engineered top-40 feature space contains strong predictive signal. Feature group ablation confirms that academic metrics (approval rates, grades) provide the primary signal (F1 = 0.671), while socioeconomic features offer complementary improvement (combined F1 = 0.717).
- t-SNE and clustering show that the three outcomes are not naturally separated, with Enrolled students lying between Dropout and Graduate.
- Among mandatory baselines, Logistic Regression is the strongest and most stable, while Decision Tree overfits and SVM benefits most from tuning.
- In open-ended exploration, tuned Random Forest achieves the best held-out test macro F1 of 0.7096, only slightly above Logistic Regression at 0.7079. Bootstrap analysis confirms this difference is not statistically significant ($p = 0.30$).
- Enrolled remains the primary challenge because it is both a minority class and a structurally transitional outcome; class weighting improves recognition but cannot fully resolve the structural overlap with Dropout and Graduate.

Limitations: The small gap between tuned RF and LR suggests that future work should prioritize better modeling of Enrolled students and external validation, rather than treating a marginal macro F1 difference as a definitive deployment advantage. Our t-SNE-based decision boundary visualizations are approximations in 2D projected space and do not represent actual high-dimensional boundaries. Clustering methods generally struggled with this dataset, likely because the discriminative features for classification do not correspond to natural density-based clusters.

Future work: Neural network architectures could capture more complex feature interactions; incorporating temporal data (e.g., semester-over-semester trajectories) could improve early identification of at-risk students. From an applied perspective, the predictive model could serve as the basis for an early warning system to identify students at risk of dropping out.

References

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[4] Claude by Anthropic (Claude Opus 4) was used for code debugging, visualization code generation, and report language polishing, accounting for less than 30% of total project effort. All core ML implementations are original student work.

Credit

Group Member Contributions

Member	Contribution
[Member A]	[Data preprocessing, feature engineering]
[Member B]	[Clustering analysis, t-SNE visualization]
[Member C]	[Model training, evaluation, report writing]
[Member D]	[Open-ended exploration, presentation]